

Market and Usability Analysis

I believe that my system is pretty good, because it's very easy to use. All players need to do is press the restart button (KEY[0]) at the very first instance of the game and at the end of every subsequent game of Connect 4. The controls are KEY[1] for moving the cursor left, KEY[2] for moving the cursor right, and KEY[3] to lock in a drop of the player's respective piece at the location of the cursor, as long as there is space in the column that the player selected the drop to be in. (If there isn't, nothing occurs, essentially forcing the player to move their cursor somewhere else for their turn.) Turns alternate automatically, so as soon as a player (P1, Red) makes a turn the system automatically begins waiting for the other player's (P2, Green) input. The first player to have 4 of their pieces in a row, whether that be horizontally, vertically, or diagonal up and right or diagonal down and right, will be evaluated as the winner immediately, "removing" the cursor and preventing any more actions in the game, while the HEX display shows whether player 1 or 2 was the winner. There are only four buttons to learn and the feedback is instant, so from an ergonomics standpoint I think it's about as approachable as a physical Connect 4 board.

In terms of cost, the entire design totals are 573 ALUTs and 130 registers, minus 30 total from clock_divider, so the total FSM cost score is 673. I'm pretty sure that its actually a pretty small design in FPGA terms so the logic itself is cheap, and the real big cost is the price of the board it has to run on. Through all of my iterations, I never experienced any glitches and my testbenches in Modelsim don't show any unwanted outputs, so I'd say that its definitely functional to a high standard. Since it's a game of Connect 4, the goal is for the game to be fun. I mean I wouldn't say that Connect 4 is a super stimulating game to begin with, but my game definitely gets the job done. Realistically, people wouldnt pay the money to buy an FPGA board that lets them play Connect 4 on an LED extension over just buying a physical board and chip. It would be more expensive and you need to play it using a power cord so it's not even that portable. As for stuff like public health, safety, and welfare, there honestly isn't much to worry about because it's a low power LED display and a few buttons, so nobody is getting hurt playing it. Environmentally it really has zero effect except for the fact that its using energy when its plugged in. However, I think the idea of playing a board game displayed digitally on a piece of hardware fits the progressing theme of digitalization and technology in the world, and socially its literally just the same game but on a new medium. Also, it was just a fun project for me to build.

User Manual

This is a connect 4 game running on the DE1-SoC. Players take turns dropping pieces into a 6x7 grid displayed on a 16x16 LED board. Player 1 is represented by red and player 2 is represented by green. The first player to line up four of their own pieces whether its horizontally, vertically, or diagonally wins and when they win it is displayed on HEX5 which player won (1 or 2). If the grid fills without any of them getting 4 in a row then the game is a draw.

The game is oriented so pieces stack upward from the bottom row so that you can tell that gravity actually exists in the game. Empty cells are off and then again red LEDs are player 1s pieces and green LEDs are player 2s pieces. A single lit LED sitting a couple of rows above the grid is the cursor which shows the current active player where in which column they are planning to drop their piece and shows whos turn it currently is because it glows in the current players color. The cursor disappears once the game ends either in a victory or a draw, and therefore the game cannot continue after it is over.

KEY[1] and KEY[2] shift the cursor left and right respectively one column toward the chosen side. The cursor stops at the edges so that it wont wrap past column 1 or column 7, while KEY[3] drops a piece into the column under the cursor. The piece animates falling from the top of the grid down to the lowest empty cell in that column, then settles. If the chosen column is already full, the drop is ignored and it stays the current players turn. Finally, KEY[0] acts as a reset and then clears the entire board, returns the cursor to the center column, and gives the turn to Player 1.

On a players turn, they need move the cursor left or right to the column they want, then press drop. Players can tell which column they are currently located on by looking at the hex display, which will show them a value 1-7. Watch the piece fall and lock into place. As soon as it lands, the system checks for a win and if there's no winner and the board isnt full, the turn automatically passes to the other player and the cursor switches to their color. Please repeat until someone wins or the board fills

When a player gets four in a row, the game freezes in a won state and the board stays displayed and the cursor turns off, so the final winning position remains visible. Same thing happens with a draw. To start a new game from either end state, press reset.

Analysis & Synthesis Resource Utilization by Entity					
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	Compilation Hierarchy Node	Combinational ALUTs	Dedicated Logic Registers	Block Memory Bits	DSP B
1	▼ DE1_SoC	573 (29)	130 (11)	0	0
1	LEDDriver:Driver	205 (205)	4 (4)	0	0
2	board_renderer:render	14 (14)	0 (0)	0	0
3	clock_divider:divider	15 (15)	15 (15)	0	0
4	▼ game_controller:game	300 (136)	100 (100)	0	0
1	win_detec...:detector	164 (164)	0 (0)	0	0
5	hex_decoder:h0	7 (7)	0 (0)	0	0
6	hex_decoder:h5	3 (3)	0 (0)	0	0